

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
2304212	Basic Telecommunication Experiments	--	2	--	--	1	--	1

Course Code	Course Name	Examination Scheme						
		Theory Marks				Term Work	Practical/ Oral	Total
		Internal assessment			End Sem. Exam			
		Test1	Test 2	Avg. of 2 Tests				
2304212	Basic Telecommunication Experiments	--	--	--	--	25	25	50

Laboratory Objectives:

1. Demonstrate the basic principles of analog and digital communication using hands-on experiments.
2. Introduce students to the use and operation of fundamental communication trainer kits and instruments.
3. Provide practical exposure to signal transmission and reception using modulation and demodulation techniques.
4. Enable students to observe and analyze the behavior of different multiplexing techniques in telecommunication.
5. Familiarize students with telephone and mobile communication basics through interactive experiments.
6. Build student confidence in interpreting real-world telecommunication systems through simulations and physical setups.

Laboratory Outcomes:

After successful completion of the course student will be able to

1. Identify basic telecommunication components and instruments such as signal generators, modulators, and demodulators.
2. Operate trainer kits to perform experiments on amplitude and frequency modulation and demodulation.
3. Demonstrate analog-to-digital and digital-to-analog signal conversions.
4. Simulate digital communication systems using audio or PC-based platforms.
5. Compare the characteristics and performance of FDM and TDM through practical setups.
6. Explain basic mobile and satellite communication setups using lab experiments.

DETAILED SYLLABUS:

Sr. No.	Module	Detailed Content	Hours	CO Mapping
0	Prerequisite	--	--	--
I	Introduction to Telecommunication Equipment	• Understanding Telephone System and Mobile Networks • Identifying Components: Routers, Switches, Modems	02	CO1
II	Signal Transmission and Reception	• Demonstration of Signal Transmission in Wired and Wireless Media • Analysis of Frequency and Bandwidth	04	CO2
III	Digital and Analog Modulation	• Experiments on AM, FM, and Digital Modulation • Visualizing Modulated Signals on an Oscilloscope	06	CO3
IV	Network Configurations and Communication	• Setting up a simple LAN and understanding IP addressing • Wi-Fi configuration and troubleshooting	06	CO4
V	Data Transmission and Error Handling	• Demonstration of Data Transmission using Serial and Parallel Ports • Experimenting with Error Detection and Correction	04	CO5
VI	Case Study and Industry Applications	• Exploring Telecom Applications in Banking, Healthcare, and E-Commerce • Group Discussions on Future Trends	04	CO6

Reference Books:

1. Frenzel, Louis E. *Principles of Electronic Communication Systems*, McGraw Hill.
2. Stallings, William. *Data and Computer Communications*, Pearson.

List of Experiments:

Week No	List of Experiments	Hrs
01	Study and Operation of a Basic Telephone System.	02
02	Identification and Demonstration of Modem, Router, and Switch Hardware.	02
03	Overview and Demonstration of Mobile Network Components.	02
04	Demonstration of Signal Transmission through Wired Media (Coaxial and Twisted Pair Cable).	02
05	Demonstration of Signal Transmission through Wireless Media (Wi-Fi or Bluetooth).	02
06	Measurement and Analysis of Signal Frequency and Bandwidth using a Spectrum Analyzer.	02

07	Generation and Observation of Amplitude Modulated (AM) Signals.	02
08	Generation and Observation of Frequency Modulated (FM) Signals.	02
09	Digital Modulation Techniques: Generation and Observation of ASK, FSK, and PSK signals	02
10	Comparison of Analog and Digital Modulated Signals using an Oscilloscope.	02
11	Setting up a Basic Local Area Network (LAN) using Switches and Routers.	02
12	Configuration and Testing of IP Addressing in a LAN setup.	02
13	Setup and Configuration of a Wi-Fi Network; Troubleshooting Common Issues.	02
14	Demonstration of Data Transmission using Serial Communication (RS-232 Standard).	02
15	Demonstration of Data Transmission using Parallel Communication Interfaces.	02
16	Implementation and Testing of Error Detection using Parity Checking and Checksums.	02
17	Case Study Presentation: Telecommunication Applications in Healthcare or E-Commerce.	04

Assessment:

Term Work: Term Work shall consist of at least 10 to 13 practicals based on the above list.

Internal Practical Exam: Conduct an internal practical exam after completing the first three modules of the course to assess and ensure the learner's understanding.

Term Work Marks: 25 Marks (Total marks) = 10 Marks (Experiment) + 10 Marks (Internal Practical Exam) + 5 Marks (Attendance)

Practical & Oral Exam: An Oral & Practical exam will be held based on the above syllabus.

Sd/-

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